

ML-J-DP4: An Integrated Quantum Mechanics-Machine Learning Approach for Ultrafast NMR Structural Elucidation

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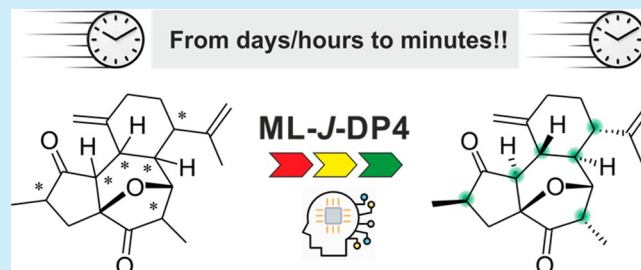


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ABSTRACT: A new tool, ML-J-DP4, provides an efficient and accurate method for determining the most likely structure of complex molecules within minutes using standard computational resources. The workflow involves combining fast Karplus-type J calculations with NMR chemical shifts predictions at the cheapest HF/STO-3G level enhanced using machine learning (ML), all embedded in the J -DP4 formalism. Our ML provides accurate predictions, which compare favorably alongside with other ML methods.



The determination of molecular structure is of crucial importance in organic, synthetic, medicinal, and natural products chemistry. The calculations of NMR chemical shifts and coupling constants using DFT approaches have emerged as powerful complementary tools to facilitate the structural assignment of complex molecules.¹ This discipline underwent an explosive growth, and experienced beneficial changes and updates over the years.^{1–4} A main breakthrough was the inclusion of sophisticated strategies to correlate the experimental and calculated data to facilitate the assignment, including Bayesian analysis,^{5–7} artificial neural network pattern recognition,⁸ parametric corrections,⁹ and conformational selection.^{7,10,11} A trend toward the use of higher levels of theory to run the NMR calculations is also noticed, positively affecting the quality of the predictions but being detrimental in terms of computational cost.³

The overall CPU time required to perform a full analysis might range from a few hours to several days (or even weeks), depending on the size and conformational flexibility of the target molecule, number of isomers under study, level of theory, and computational resources availability. Therefore, the development of faster strategies that allow a preliminary and safe assignment within a few minutes is essential to speed up the *in silico* elucidation process. In this letter we wish to report our proof-of-principle machine-learning based approach that allows acceleration of J -DP4 (one of the state-of-the-art methods in structural elucidation) by 2 orders of magnitude.

J -DP4 is a modification of DP4 that exploits scalar $^3J_{\text{HH}}$ couplings to constrain the conformational sampling (iJ -DP4) and to provide important stereochemical information through the corresponding Bayesian probability term (dJ -DP4).⁷ In the trade off between accuracy and cost, J -DP4 emerges as one of

the leading methods, capable of solving highly complex structures significantly faster than other alternatives.¹²

Our strategy to accelerate J -DP4 was based on two pillars: replacing the expensive DFT-calculated $^3J_{\text{HH}}$ values with lighter approximations, and computing the isotropic shielding constants (σ) with faster methods. With respect to the first question, the Karplus-type equations proved to be ideal targets, as they provide a cosine-modulated relationship between the dihedral angle of the coupled nuclei (φ) and their $^3J_{\text{HH}}$ coupling, making the calculation process almost instantaneous. To investigate our working hypothesis, we chose two approaches to estimate the $^3J_{\text{HH}}$ values: one of the variants of the well-known Karplus equation, and the general Haasnoot–de Leeuw–Altona (HLA) equation (see [Supporting Information](#) (SI)). The first was chosen for its simplicity, while the second is one of the most widely used equations for predicting the scalar coupling of organic molecules.¹³ In view of the second question, the σ calculations were investigated at the HF/3-21G, HF/MIDI! and HF/STO-3G levels, which are undoubtedly the lowest levels at which GIAO NMR calculations can be performed. Since only modest predictions could be expected at these levels, we envisioned a machine learning to improve the quality of the results at negligible additional cost ([Figure 1](#)).

A test set of 25 challenging examples was created, showing an interesting diversity of molecular complexity and conforma-

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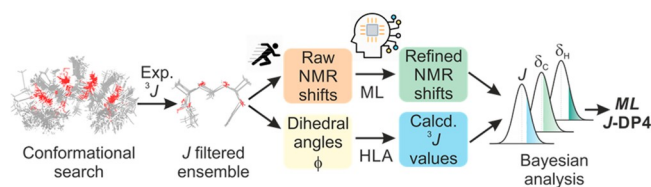


Figure 1. Representation of the ML-J-DP4 method.

tional freedom (see SI). The nonquantum methods for estimating $^3J_{\text{HH}}$ provided acceptable agreement with the results computed at the B3LYP/6-31G** level (the standard in J-DP4), with R^2 values of 0.88 and 0.91 for Karplus and HLA, respectively (see SI). To quantify the performance of each method, we developed a ranking system that weights the calculated probability of the correct isomer, and the relative order in which that structure was assigned (see SI). Figure 2 shows the results obtained for both dJ-DP4 and iJ/dJ-DP4.

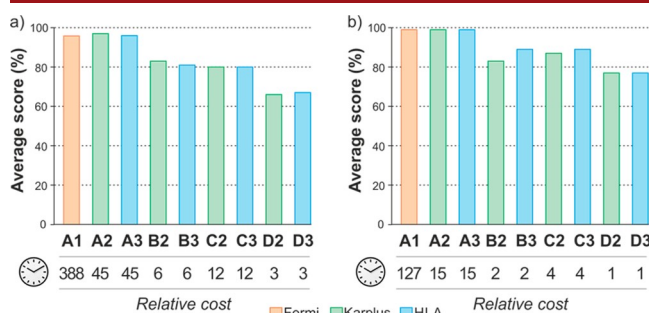


Figure 2. Overall performance of dJ-DP4 (Figure 2a) and iJ/dJ-DP4 (Figure 2b) methods computed for the test set with the NMR chemical shifts calculated at the B3LYP/6-31G**//MMFF (A), HF/3-21G//MMFF (B), HF/MIDI1//MMFF (C), and HF/STO-3G//MMFF (D) levels, and the 3J coupling constants calculated at the corresponding levels using the Fermi contact (method A1), the Karplus equation (methods A2, B2, C2, and D2, respectively), and the HLA equation (methods A3, B3, C3, and D3, respectively).

The results obtained with A1, A2, and A3 provided strong evidence for the inclusion of Karplus-type $^3J_{\text{HH}}$ values in the J-DP4 architecture, resulting in significant time savings (*vide infra*). In analyzing the impact of the theory level for performing the NMR calculations, we observed an expected degradation of the classification capacity as a result of the poor NMR prediction. In the case of level A, the mean absolute errors are 1.61 and 0.13 ppm for carbon and proton data, respectively, which is in agreement with previous reports.⁵ However, the accuracy decreased significantly when the theory level was lowered, with values of 3.71 ppm/0.24 ppm (level B), 3.81 ppm/0.24 ppm (level C), and 6.00 ppm/0.42 ppm (level D). These results are consistent with previous findings and demonstrate that low quality basis sets (mainly the STO-3G) are unsuitable for accurate NMR calculations.¹⁴

The results obtained with iJ/dJ-DP4 (Figure 2b) were slightly better than the corresponding dJ-DP4 values (Figure 2a), but with a remarkable reduction in the computational cost. Using only a few $^3J_{\text{HH}}$ couplings to constrain the conformational search (typically between 1 and 3), the total number of conformations was reduced by 60% on average (from 13% to 91%). In this regard, the effect of replacing quantum-computed $^3J_{\text{HH}}$ with Karplus-type values is evident, with an ~ 10 -fold reduction in average cost. In addition, the effect of the level of

theory used to calculate the NMR chemical shifts is also notable, with ~ 5.3 - (level B), ~ 2.9 - (level C), and ~ 11.1 -fold (level D) reductions (Figure 2).

In an attempt to obtain the classification capacity of level A at the computational cost of level D, we envisaged the use of ML. Exciting work has been done in the area of NMR chemical shift prediction by translating 3D molecular descriptors with the aid of ML. Some key publications are summarized in the SI. In these cases, the local environments were encoded with a single representation, such as Coulomb Matrix,¹⁵ SOAP,¹⁶ FCHL,¹⁷ 3D-DensNet,¹⁸ or GNN.¹⁹ The results obtained with these MLs are remarkable, especially considering that the descriptors are not derived from DFT calculations. The accuracy levels obtained (relative to DFT) are clearly sufficient to distinguish between constitutional isomers, but the situation might be different when dealing with diastereoisomers that exhibit subtle differences in their chemical environments. Although some of these MLs have been successfully evaluated to tackle stereoisomerism,^{17,19} in our opinion the small number of examples provided prevents a thorough evaluation of their actual usefulness in this field. To illustrate how difficult is to distinguish closely related structures using ML-computed NMR chemical shifts, we evaluated some of the compounds included in our test set using CASCADE, one of the most accurate new methods that also offers a user-friendly platform.¹⁹ As can be seen in the SI, the results obtained with DP4 were modest, with some compounds being misassigned. This should in no way be taken as a criticism of this cutting edge method, but rather highlights the complexity associated with stereochemical mapping of similar structures.

Beran and co-workers recently demonstrated that highly accurate chemical shifts can be obtained through a Δ -ML approach, in which an ensemble of neural networks was trained to correct PBE0/6-31G// ω B97XD chemical shifts to the target PBE0/6-311+G(2d,p)// ω B97XD by using the atomic environment vector (AEV) to encode the geometric information on the nuclear environment.²⁰ The RMS values reached were only 0.11 and 0.70 ppm for the ^1H and ^{13}C NMR chemical shifts, respectively, which are smaller than those presented in Table S4. However, when using minimal basis sets (STO-3G) in the low level, the RMSs were remarkably higher (0.215 and 2.5 ppm for ^1H and ^{13}C data, respectively). Our strategy borrows from this approach, but explores a hybrid representation of the molecular environments, including isotropic shielding constants computed at the cheapest HF/STO-3G//MMFF level coupled with other local descriptors easily obtained from the corresponding 3D molecular structure.

Our ML workflow is summarized in Figure 3 and described in detail in the SI. After selecting the 17,000 most diverse structures from a full set of 170,000, the 3D geometries were extracted from the SMILES representation of each molecule followed by geometry optimization with the MMFF force field, culminating in NMR calculations at the HF/STO-3G and B3LYP/6-31G** levels (low and high, respectively). In this proof-of-concept we used B3LYP/6-31G** as the high level to reproduce the results obtained with the standard J-DP4 method. However, this approach could be performed with any desired level of theory. The low level NMR calculations were carried out using the pop = nbo option (very light in terms of computational cost) to compute potentially useful parameters, such as hybridization. The data were randomly divided into 17,000 molecules for training (232,560 ^{13}C and

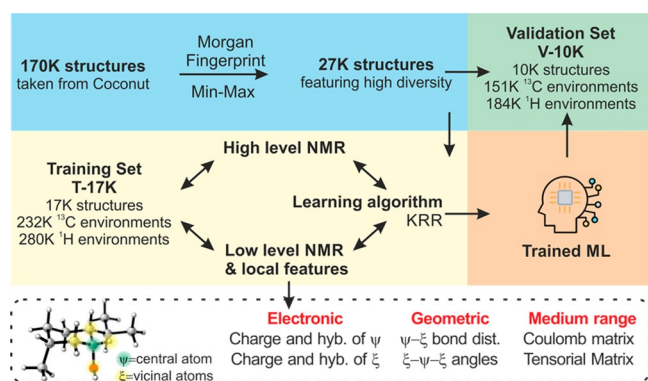


Figure 3. Scheme of the ML used for improving the HF/STO-3G chemical shifts predictions.

280,446 ^1H values, T17k set) and 10,000 molecules for validation (150,760 ^{13}C and 183,612 ^1H values, V10k set). The whole process was automated by homemade Python scripts.

To capture the local environments of the various atoms, we considered different geometric and electronic descriptors of the central atom (including paramagnetic and diamagnetic contributions of σ , charge and hybridization), vicinal atoms (atomic number, charge, hybridization, distances, and bond angles), and long-range interactions (Coulomb matrix and the Tensorial matrix defined here); see SI for more details. Our ML strategy was based on a Kernel Ridge Regression (KRR) to refine the HF/STO-3G shielding values. We also tried other common approaches, such as Supported Vector Machine (SVM), K-nearest neighbor (KNN), and LASSO following the adaptive training discussed later, but the best results were obtained with KRR, a nonlinear form of regression with regularization to prevent overfitting.²¹

In KRR strategies, the size of the training matrix has a significant impact on the total computational cost. For example, in recent ML approaches for calculating NMR properties the number of environments is in the range 15k–35k.^{16,17,22} In previous works, the training matrices were generated using data from structures selected by random,^{16,19,22} or adaptive strategies.¹⁷ This implies that similar environments could in principle be shared by two or more structures, or in other words, that a molecule could have redundant or unhelpful environments. Because NMR properties are local in nature, we did not consider it mandatory to take all the environments of a test molecule, but only the most important ones. Therefore, our selection strategy was based on finding the best combination of individual environments (rather than individual molecules) that maximizes the predictive capacity of the ML. The selection of appropriate descriptors is also important, as some of them may behave erratically, or interact negatively with other descriptors. Overall, the best combination of descriptors was the E767 ensemble (see SI for more details) for both ^{13}C and ^1H values, suggesting that this selection encodes well the similarity between chemical environments. To improve the accuracy of ML, we used two adaptive learning strategies. One was based on a modification of the variance approach implemented in IMPRESSION,¹⁷ but using single data points rather than whole molecules to create the optimal training set. The other adaptive learning strategy was similar, but instead of ranking the environments by variance, the selection was based on the

largest errors. In this way, the 1,000 examples of the validation set that had the largest errors were added to the training set in each iteration. In this respect, we speculated that the environments that were poorly described by the original training set were incorporated to improve the prediction in the next stage. As described in the SI, the best results were obtained with this last approach (MAE 1.19 ppm vs 1.53 ppm for carbon data, and 0.13 ppm vs 0.16 ppm for proton data). The hyperparameters of the resulting machines were further optimized with a 5-fold approach, and the optimal machines were tested against the independent test set (V10k, 150,760 ^{13}C and 183,612 ^1H values). The predictions were highly satisfactory, with MAE of 1.21 ppm/0.14 ppm, RMS of 1.63 ppm/0.19 ppm, and MaxErr of 20.74 ppm/1.89 ppm for ^{13}C and ^1H data, respectively (see SI). These results were excellent compared with those obtained from other recent ML approaches discussed above. While it is true that our ML requires quantum computation of isotropic shieldings, from our point of view the quality of the results justifies this extremely small additional cost.

With the ML trained, we next recomputed the *J*-DP4 values of our test set (Figure 4a). To our delight, we obtained similar

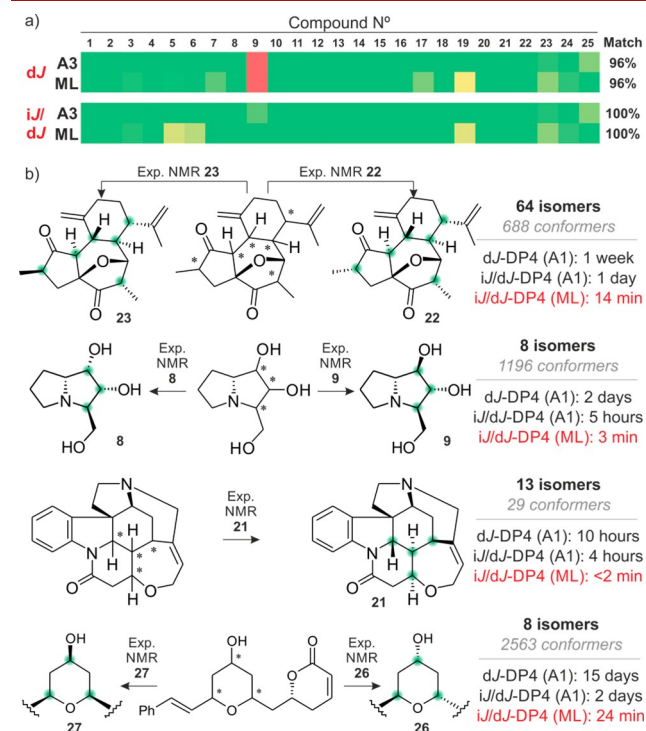


Figure 4. (a) Overall performance of ML-J- DP4. (b) Representative case studies to illustrate the power of ML-J- DP4.

results as with the A-level, but at a fraction of the computational cost. Considering that post-training calculations of ML take seconds to minutes depending on the system, the total cost of the ML-J- DP4 is almost the same as the D level. The fact that complex molecular architectures, such as curcusones 22 and 23, hyacinthacines 8 and 9, and strychnine (21) could be solved in such a short time using a single office computer represents a meritorious achievement.

To further demonstrate the utility of the present method, we included 7 additional examples, including the revised and putative structures of cryptoconatone H (compounds 26 and

27, Figure 4b),²³ representative examples of flexible 5- and 6-membered rings, and flabelliparicine, a complex monoterpene bisindole alkaloid.²⁴ In all cases, the correct isomer was identified with high probability, requiring only a few minutes of computational work. It is important to emphasize that this proof-of-concept ML was trained with molecules that contained first-row atoms (a common strategy followed in other ML approaches),¹⁷ which could limit the range of chemical space. However, we tested molecules containing second-row or heavier atoms (including S, Cl, and Br), and good results were obtained after removing the conflicting resonances. Finally, we tested ML-J-DP4 in the stereochemical determination of belizentrin methyl ester, a molecule that is extremely challenging for *in silico* elucidation.¹² Using a fragment approach, the correct isomer (among 32,168 candidates) was successfully determined in only ~90 min using a single desktop computer (see SI for more details). We started by determining the most likely relative configuration of the 3 stereoclusters F1, F2, and F3 (Figure 5), demanding less

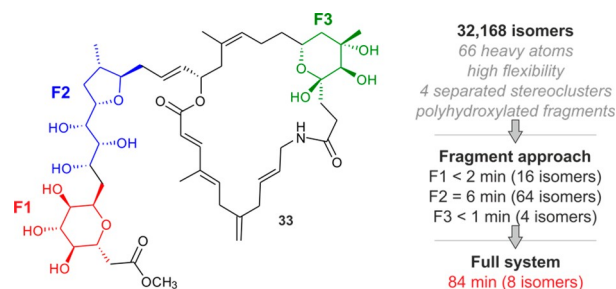


Figure 5. Performance of ML-J-DP4 in the stereochemical elucidation of belizentrin methyl ester.

than 10 min. Finally, we connected them to settle the relative configuration of the full molecule analyzing the eight possible diastereoisomers. To better appreciate the magnitude of the results obtained, it is important to note that the elucidation of 33 would have required months with the standard DP4 method using comparable computational resources.¹²

In summary, by combining *J* calculations using Karplus-type equations with NMR chemical shifts calculations using the cheapest HF/STO-3G method, further refined by machine learning, we have developed an ultrafast methodology that allows the assignment of complex systems in a few minutes using standard computational resources. The degree of speed-up achieved is remarkable compared to other state-of-the-art methods, making this an excellent approach to use at early stages in the discovery of new natural or synthetic products. The entire process was automated in the form of a Python script and released under an open source MIT license. It is available at https://github.com/Sarotti-Lab/ML_J_DP4 (including instructions and tutorials) and from the Python Package Index (<https://pypi.org/project/ml-jdp4/>).

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.2c01251>.

Computational details, structures of all compounds under study, NMR data and probability results, and details on the ML training and validation (PDF)

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Notes

The authors declare no competing financial interest.

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